

MadhuSai Pathakoti

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TECHNICAL SKILLS

Programming Languages: Python, SQL

Databases: MongoDB, MySQL, PostgreSQL, Redis

Backend & APIs: FastAPI, Flask, REST APIs, Microservices Architecture

DevOps: Docker, Git, Gitlab

Data & ML: NumPy, Pandas, Scikit-learn, Time-Series Forecasting, TensorFlow

Data Visualization & BI: Google Looker, Looker Studio

Core CS: Data Structures, Algorithms, Object-Oriented Design, Design Patterns, LLD, HLD, Performance Optimization

PROFESSIONAL EXPERIENCE

Software Engineer

Sep 2024 – Present

Enmovil IT Solutions

Hyderabad, India

- Architected scalable forecasting platform supporting ML models (SARIMA, VAR, Prophet, XGBoost, LSTM) with systematized model selection based on time-series characteristics, serving 7+ enterprise clients
- Built 45+ high-throughput FastAPI endpoints for training, inference, and experiment management with robust error handling and validation, supporting 1000+ daily API calls
- Implemented computerized data pipelines for preprocessing noisy enterprise datasets, improving forecast accuracy by 15-20% through feature engineering and outlier detection
- Planned enterprise inventory optimization system spanning multi-tier warehouses with algorithms for safety stock calculation, fair-share allocation, and batch-aware placement using freshness constraints

Junior Backend Developer

Jun 2024 – Aug 2024

Enmovil IT Solutions

Hyderabad, India

- Engineered 3D bin-packing optimization engine maximizing container utilization across 6 spatial orientations with constraint-driven algorithms for fragile items, weight distribution, and priority sequencing
- Optimized computational geometry paths using vectorized NumPy operations, reducing planning latency from 120-180s to 5-15s (24x speedup) through algorithmic improvements and caching strategies
- Developed 25+ RESTful APIs in Flask for real-time container orchestration with MongoDB-backed configuration management, enabling zero-downtime rule updates
- Built 3D visualization tooling for packing validation and debugging, reducing QA time by 40%

KEY PROJECTS

[Distributed Task Queue System](#) | *FastAPI, Redis, PostgreSQL, Python, Render*

- Created distributed task queue supporting priority-based scheduling, exponential backoff retry, and dead letter queue — deployed live with real-time monitoring dashboard
- Developed Redis Sorted Set based priority queue enabling $O(\log N)$ task insertion and $O(1)$ retrieval, with atomic SET NX EX worker locking to prevent duplicate task processing across parallel workers
- Built fault-tolerant retry system with exponential backoff (2s $\rightarrow$ 4s $\rightarrow$ 8s) and dead letter queue for tasks exhausting all retry attempts, ensuring zero task loss on worker failures
- Developed real-time monitoring dashboard with live task status tracking, metrics visualization, and task detail inspection — structured using SOLID principles for maintainability and extensibility

[Patient Discharge Decision Intelligence System](#) | *FastAPI, XGBoost, Rule Engine, Streamlit*

- Built healthcare Decision Intelligence platform combining ML models with clinical safety rules for patient discharge predictions with full audit trails and SHAP-based explainability
- Implemented rule engine architecture with priority-based evaluation and model governance framework ensuring clinical safety constraints are never violated
- Developed using SOLID principles to ensure maintainability, extensibility, and production readiness

EDUCATION

Rajiv Gandhi University of Knowledge Technologies (RGUKT)

Bachelor of Technology in Electronics and Communication Engineering

Basar, Telangana

Sep 2020 – Apr 2024